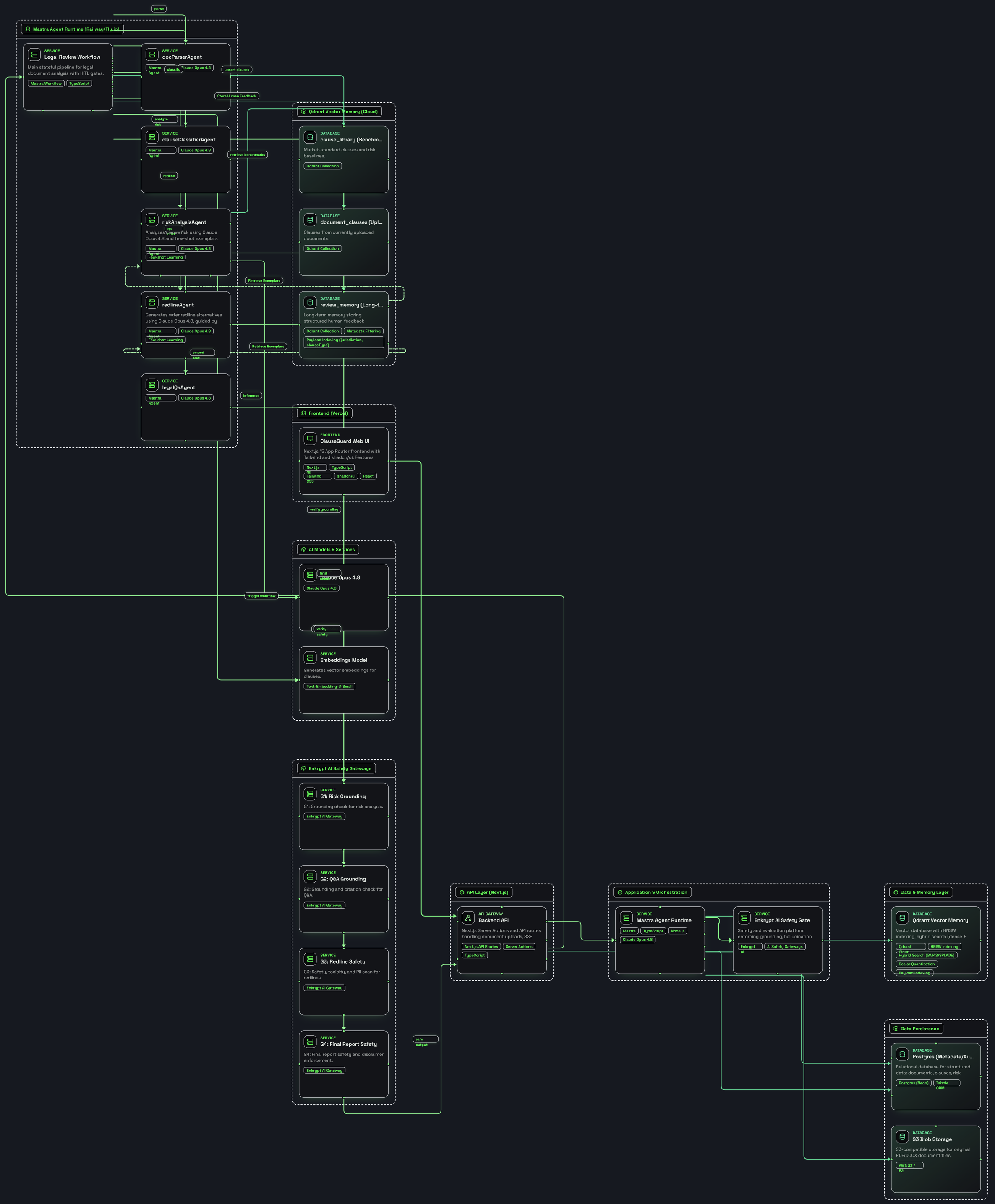
parse
Mastra Agent Runtime (Railway/Fly.io)
SERVICE
Legal Review Workflow
Main stateful pipeline for legal document analysis with HITL gates.
Mastra Workflow
TypeScript
SERVICE
docParserAgent
Mastra Agent
classify
Claude Opus 4.8
upsert clauses
Store Human Feedback
Qdrant Vector Memory (Cloud)
analyze risk
SERVICE
clauseClassifierAgent
Mastra Agent
Claude Opus 4.8
retrieve benchmarks
redline
DATABASE
clause_library (Benchm...
Market-standard clauses and risk baselines.
Qdrant Collection
SERVICE
riskAnalysisAgent
qa chat
Analyzes clause risk using Claude Opus 4.8 and few-shot exemplars
Mastra Agent
Claude Opus 4.8
Few-shot Learning
DATABASE
document_clauses (Upl...
Clauses from currently uploaded documents.
Qdrant Collection
Retrieve Exemplars
SERVICE
redlineAgent
Generates safer redline alternatives using Claude Opus 4.8, guided by
Mastra Agent
Claude Opus 4.8
Few-shot Learning
embed text
DATABASE
review_memory (Long-t...
Long-term memory storing structured human feedback
Qdrant Collection
Metadata Filtering
Payload Indexing (jurisdiction, clauseType)
Retrieve Exemplars
SERVICE
legalQaAgent
Mastra Agent
Claude Opus 4.8
inference
Frontend (Vercel)
FRONTEND
ClauseGuard Web UI
Next.js 15 App Router frontend with Tailwind and shadcn/ui. Features
Next.js 15
TypeScript
Tailwind CSS
shadcn/ui
React
verify grounding
AI Models & Services
final answer
Claude Opus 4.8
Claude Opus 4.8
trigger workflow
verify safety
SERVICE
Embeddings Model
Generates vector embeddings for clauses.
Text-Embedding-3-Small
Enkrypt AI Safety Gateways
SERVICE
G1: Risk Grounding
G1: Grounding check for risk analysis.
Enkrypt AI Gateway
SERVICE
G2: Q&A Grounding
G2: Grounding and citation check for Q&A.
Enkrypt AI Gateway
SERVICE
G3: Redline Safety
G3: Safety, toxicity, and PII scan for redlines.
Enkrypt AI Gateway
SERVICE
G4: Final Report Safety
G4: Final report safety and disclaimer enforcement.
Enkrypt AI Gateway
safe output
API Layer (Next.js)
API GATEWAY
Backend API
Next.js Server Actions and API routes handling document uploads, SSE
Next.js API Routes
Server Actions
TypeScript
Application & Orchestration
SERVICE
Mastra Agent Runtime
Mastra
TypeScript
Node.js
Claude Opus 4.8
SERVICE
Enkrypt AI Safety Gate
Safety and evaluation platform enforcing grounding, hallucination
Enkrypt AI
AI Safety Gateways
Data & Memory Layer
DATABASE
Qdrant Vector Memory
Vector database with HNSW indexing, hybrid search (dense +
Qdrant Cloud
HNSW Indexing
Hybrid Search (BM42/SPLADE)
Scalar Quantization
Payload Indexing
Data Persistence
DATABASE
Postgres (Metadata/Au...
Relational database for structured data: documents, clauses, risk
Postgres (Neon)
Drizzle ORM
DATABASE
S3 Blob Storage
S3-compatible storage for original PDF/DOCX document files.
AWS S3 / R2